DATA 622 Homework 4 Redo: Regularization and Model Selection

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1 Assignment Setup

	quantity	value
0	stints	32358
1	player coefficients	539
2	games	1225

This report uses leave-one-out cross-validation for the OLS baseline, includes the unweighted `r_unw` ranking table and RAPM-versus-minutes plot, corrects the weighting diagnostic, discusses how the rankings change after weighting, and plots uncertainty versus minutes for the full player pool.

2 Problem 1: Ridge Regression for Inference

2.1 Part (a): Ordinary Linear Regression

For OLS, leave-one-out cross-validation is especially convenient because the fitted-value leverages let us compute the LOOCV error directly from one model fit.

	metric	value
0	OLS in-sample RMSE	69.6289
1	OLS LOOCV RMSE	70.7875
2	LOOCV - train gap	1.1586

The LOOCV RMSE is larger than the training RMSE, so the OLS fit does lose some accuracy out of sample. The gap is noticeable but not extreme, which makes OLS a reasonable baseline but not the model I would lean on most for RAPM.

Interpretation

That pattern makes sense for this problem. The design matrix is wide and highly collinear because players share the floor in many overlapping lineups. OLS can still fit the data, but a regularized model should be more stable than an unpenalized one.

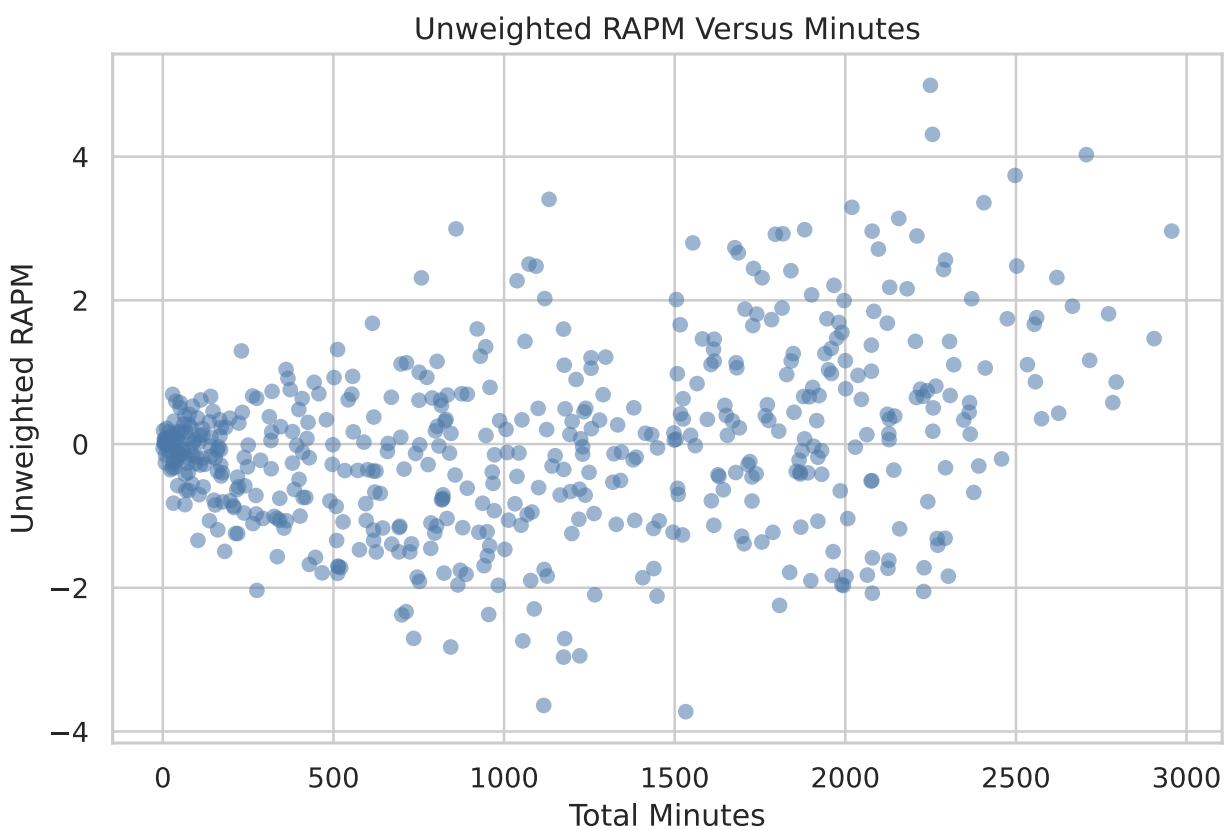
2.2 Part (b): Examining RAPM Coefficients

I fit ridge regression on the unweighted outcome using the required alpha grid and choose the penalty with an efficient generalized cross-validation calculation on the lineup matrix.

	metric	value
0	selected alpha	908.5176
1	ridge in-sample RMSE	70.0791
2	OLS LOOCV RMSE for comparison	70.7875
3	OLS to ridge train RMSE difference	-0.4502

	alpha	gcv	df_eff
17	908.51758	0.15320	150.58388
16	464.15888	0.15323	212.92644
18	1778.27941	0.15328	99.50668
15	237.13737	0.15340	279.61801
19	3480.70059	0.15342	62.09004
20	6812.92069	0.15356	36.84851
14	121.15277	0.15366	342.44108
21	13335.21432	0.15368	20.89356

	r_unw	player_name	player_id	rapm_unweighted	minutes
0	1	Joel Embiid	203954	4.992	2249.47
1	2	Nikola Jokic	203999	4.310	2255.54
2	3	Trae Young	1629027	4.028	2706.40
3	4	Pascal Siakam	1627783	3.738	2497.48
4	5	Kevin Love	201567	3.405	1131.82
5	6	Jalen Brunson	1628973	3.359	2406.14
6	7	Draymond Green	203110	3.294	2019.37
7	8	Brook Lopez	201572	3.140	2156.65
8	9	Zion Williamson	1629627	2.994	859.13
9	10	Anthony Davis	203076	2.983	1880.81
10	11	Julius Randle	203944	2.964	2956.55
11	12	Derrick White	1628401	2.963	2078.65
12	13	Kawhi Leonard	202695	2.927	1817.22
13	14	Coby White	1629632	2.919	1795.01
14	15	Darius Garland	1629636	2.896	2209.82
15	16	Isaiah Joe	1630198	2.800	1553.19
16	17	Myles Turner	1626167	2.733	1676.32
17	18	Jrue Holiday	201950	2.713	2097.00
18	19	Michael Porter Jr.	1629008	2.661	1686.43
19	20	Franz Wagner	1630532	2.561	2293.49



The table is sorted by `r_unw`. The top unweighted RAPM player is Joel Embiid, followed by Nikola Jokic and Trae Young.

Interpretation

The scatterplot shows why minutes matter for interpretation. Strong RAPM values from players with larger minute totals are easier to trust than equally large values from players with smaller samples. The names near the top of the list are also a useful basketball sanity check: seeing players such as Joel Embiid and Nikola Jokic near the top makes the model feel more plausible than if the list were dominated only by low-minute role players.

2.3 Part (c): Ridge RAPM

	top_10_unweighted_players
0	Joel Embiid
1	Nikola Jokic
2	Trae Young
3	Pascal Siakam
4	Kevin Love
5	Jalen Brunson
6	Draymond Green
7	Brook Lopez
8	Zion Williamson
9	Anthony Davis

The top of the unweighted ridge list includes several players who pass a basic basketball reality check, along with a few names that are harder to believe at face value. That is typical for a single-season RAPM model because it is trying to separate player effects from noisy lineup combinations.

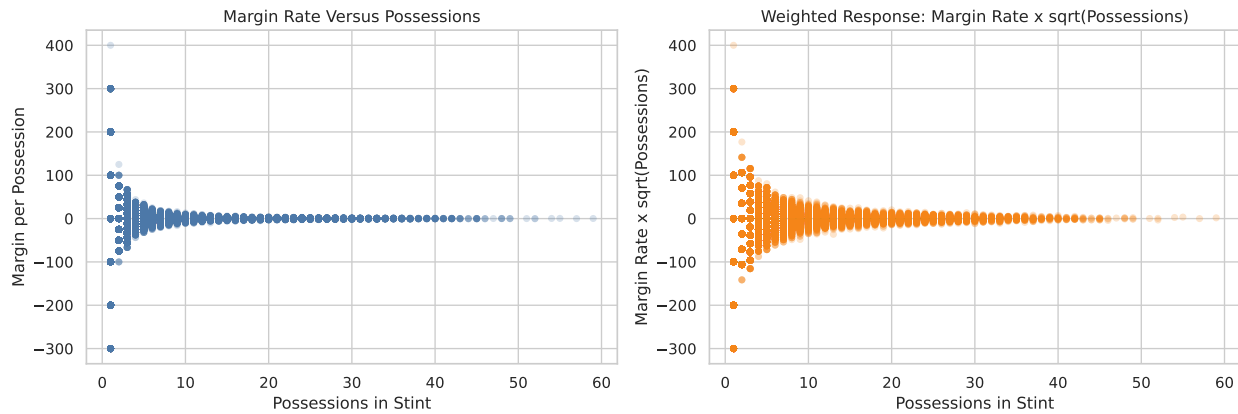
Interpretation

The rankings are believable in a broad sense, but I would not read them as exact truths. A player can look better or worse depending on teammates, opponents, minute totals, and how much regularization pulls noisy estimates back toward zero. That is why the minutes table in part (b) matters: it helps separate strong estimates with a real sample behind them from results that may be driven more by noise.

3 Problem 2: Possession Weights

3.1 Part (a): Heteroscedasticity in Stints

The key object is the margin per possession, `margin_rate = margin / n_pos`. Short stints should have noisier rates because each possession has more influence on the observed average.



The first plot shows a funnel shape: low-possession stints have much more variable margin rates. In the second plot, multiplying the response by `sqrt(n_pos)` makes the spread much more stable across possession levels, which is the pattern we want if weighting by possessions is correcting heteroscedasticity rather than making it worse.

Interpretation

This is the main statistical reason to weight the regression by `n_pos`. Longer stints contain more information about lineup performance than extremely short stints, so they should have more influence in estimation.

3.2 Part (b): Implementing Weighted Regression

	metric	value
0	weighted RMSE using n_pos weights	45.2549
1	unweighted RMSE from weighted model predictions	70.1234

	player_name	r_unw	r_w	shift_w_minus_unw	rapm_unweighted	rapm_weighted	minutes
0	Joel Embiid	1	4	3	4.992	5.670	2249.47
1	Nikola Jokic	2	1	-1	4.310	6.595	2255.54
2	Trae Young	3	63	60	4.028	2.409	2706.40
3	Pascal Siakam	4	57	53	3.738	2.538	2497.48
4	Kevin Love	5	170	165	3.405	0.848	1131.82
5	Jalen Brunson	6	146	140	3.359	1.051	2406.14
6	Draymond Green	7	2	-5	3.294	6.008	2019.37
7	Brook Lopez	8	16	8	3.140	3.613	2156.65
8	Zion Williamson	9	9	0	2.994	4.849	859.13
9	Anthony Davis	10	3	-7	2.983	5.816	1880.81
10	Julius Randle	11	52	41	2.964	2.672	2956.55
11	Derrick White	12	43	31	2.963	2.917	2078.65
12	Kawhi Leonard	13	7	-6	2.927	5.134	1817.22
13	Coby White	14	72	58	2.919	2.292	1795.01
14	Darius Garland	15	40	25	2.896	2.931	2209.82
15	Isaiah Joe	16	124	108	2.800	1.377	1553.19
16	Myles Turner	17	29	12	2.733	3.118	1676.32

	player_name	r_unw	r_w	shift_w_minus_unw	rapm_unweighted	rapm_weighted	minutes
17	Jrue Holiday	18	10	-8	2.713	4.741	2097.00
18	Michael Porter Jr.	19	37	18	2.661	2.984	1686.43
19	Franz Wagner	20	24	4	2.561	3.316	2293.49

	player_name	r_unw	r_w	shift_w_minus_unw	minutes
467	Clint Capela	468	49	-419	1696.07
456	Jonathan Isaac	457	109	-348	160.89
464	Matisse Thybulle	465	118	-347	1198.67
517	Deni Avdija	518	186	-332	1989.40
439	Josh Green	440	113	-327	1454.54

	player_name	r_unw	r_w	shift_w_minus_unw	minutes
97	Malik Beasley	98	471	373	2037.12
80	Norman Powell	81	429	348	1679.88
188	Lonnie Walker IV	189	533	344	1199.52
143	Jordan Poole	144	473	329	2364.07
154	Jake LaRavia	155	476	321	398.53

Relative to the unweighted model, Clint Capela is one of the biggest movers upward after weighting, while Malik Beasley falls the most. The weighted top 10 are Nikola Jokic, Draymond Green, Anthony Davis, Joel Embiid, Christian Koloko, Boban Marjanovic, Kawhi Leonard, Giannis Antetokounmpo, Zion Williamson, Jrue Holiday.

Interpretation

The ranking changes are the main takeaway in Problem 2. Weighting gives more influence to high-possession stints, so players whose unweighted RAPM was driven more by short, volatile samples tend to lose ground, while players with stronger results over larger possession samples tend to rise or stay near the top. The weighted rankings look more conservative and more believable.

4 Problem 3: Interpreting Bootstrap Uncertainty

4.1 Part (a): Calculate Confidence Intervals Using the Bootstrap

I bootstrap stint rows, refit the weighted ridge model each time, and keep the full coefficient vector so that uncertainty can be examined for all players, not just the top 20.

	player_name	rapm_weighted	boot_se	ci92_low	ci92_high	minutes
1	Nikola Jokic	6.595	1.356	4.541	7.663	2255.54
6	Draymond Green	6.008	2.829	4.189	10.363	2019.37
9	Anthony Davis	5.816	2.732	3.427	9.873	1880.81
0	Joel Embiid	5.670	0.893	4.234	6.283	2249.47
28	Christian Koloko	5.286	2.680	1.877	7.954	757.82
178	Boban Marjanovic	5.208	1.304	5.230	8.163	166.57

	player_name	rapm_weighted	boot_se	ci92_low	ci92_high	minutes
12	Kawhi Leonard	5.134	1.221	2.646	5.189	1817.22
25	Giannis Antetokounmpo	4.900	1.425	2.438	5.639	1840.59
8	Zion Williamson	4.849	1.326	3.883	6.703	859.13
17	Jrue Holiday	4.741	0.706	4.384	5.911	2097.00
30	Aaron Gordon	3.974	1.598	2.501	6.335	1966.73
197	Jared Rhoden	3.940	2.285	-0.378	4.926	184.23
20	Cameron Johnson	3.740	0.780	2.959	4.735	1073.00
46	Josh Hart	3.677	0.780	1.920	3.620	2474.72
98	Charles Bassey	3.614	2.444	-1.138	4.611	556.20
7	Brook Lopez	3.613	1.516	2.053	5.531	2156.65
218	Zach Collins	3.533	0.722	3.216	4.883	1414.20
233	Chimezie Metu	3.466	2.422	0.680	6.330	697.41
69	Kira Lewis Jr.	3.447	1.698	0.025	3.785	230.24
42	Jayson Tatum	3.432	2.229	2.087	6.629	2771.44

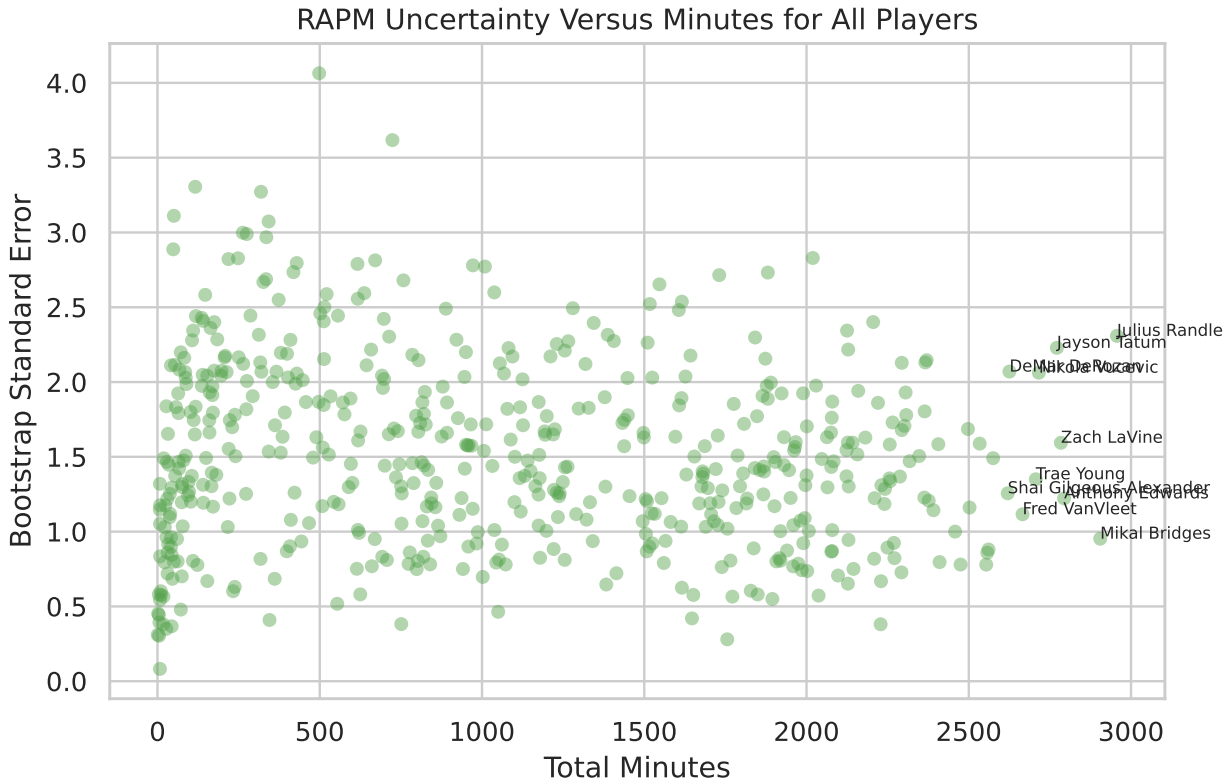
	pairwise overlap rate among top-20 92% intervals
0	0.911

There is substantial overlap across many of the top players' 92% intervals, which means the ordering at the very top should not be treated as perfectly precise.

Interpretation

The top of a RAPM ranking usually looks more precise than it really is. The bootstrap intervals show that some players are clearly positive, but many nearby ranks are hard to separate in a meaningful way.

4.2 Part (b): Uncertainty Versus Minutes



The plot now includes all players. In general, uncertainty falls as minutes increase, but the pattern is not perfectly smooth because RAPM is also affected by regularization and lineup collinearity.

Interpretation

This full-player plot is more informative than looking only at the top 20 because it shows the overall pattern directly. Low-minute players tend to have much noisier RAPM estimates, while heavy-minute players are usually estimated with more precision.

4.3 Part (c): Bayesian-Style Comparison

	player_id	player_name	boot_se	bayes_se	abs_diff
6	203110	Draymond Green	2.829	1.635	1.194
9	203076	Anthony Davis	2.732	1.612	1.120
20	1629661	Cameron Johnson	0.780	1.774	0.994
218	1628380	Zach Collins	0.722	1.646	0.924
17	201950	Jrue Holiday	0.706	1.596	0.889
28	1631132	Christian Koloko	2.680	1.827	0.853
0	203954	Joel Embiid	0.893	1.630	0.737
178	1626246	Boban Marjanovic	1.304	2.004	0.700
46	1628404	Josh Hart	0.780	1.467	0.687
42	1628369	Jayson Tatum	2.229	1.576	0.652
233	1629002	Chimezie Metu	2.422	1.812	0.610

	player_id	player_name	boot_se	bayes_se	abs_diff
98	1629646	Charles Bassey	2.444	1.910	0.533
8	1629627	Zion Williamson	1.326	1.781	0.456
12	202695	Kawhi Leonard	1.221	1.594	0.373
69	1630184	Kira Lewis Jr.	1.698	2.059	0.361
1	203999	Nikola Jokic	1.356	1.711	0.355
197	1631197	Jared Rhoden	2.285	2.041	0.244
25	203507	Giannis Antetokounmpo	1.425	1.567	0.142
30	203932	Aaron Gordon	1.598	1.739	0.140
7	201572	Brook Lopez	1.516	1.561	0.045

The bootstrap and Bayesian-style standard errors are broadly similar for many players, but the biggest disagreements occur for the noisier cases.

Interpretation

That comparison makes sense to me. The Bayesian-style formula is cleaner and faster, but it leans more heavily on the model assumptions. The bootstrap is slower, yet it reflects empirical instability more directly, which is especially useful for players whose estimates move around when the stints are resampled.

5 Extra Credit: Ridge Versus Lasso and Interval Caveat

5.1 Part (a): Ridge Versus Lasso

For the extra credit, I fit a lasso model on the same unweighted design matrix and compare it to the ridge fit from Problem 1. I use a separate alpha grid for lasso, following the scale suggested in the lab.

	metric	value
0	selected lasso alpha	0.0695
1	nonzero lasso coefficients	105.0000
2	share of coefficients set to zero	0.8052

	player_name	ridge_rapm	lasso_rapm	minutes
156	Joel Embiid	4.992	9.392	2249.47
258	Nikola Jokic	4.310	8.170	2255.54
48	Trae Young	4.028	5.193	2706.40
99	Pascal Siakam	3.738	6.224	2497.48
318	Kevin Love	3.405	5.715	1131.82
218	Jalen Brunson	3.359	3.389	2406.14
2	Draymond Green	3.294	4.547	2019.37
242	Brook Lopez	3.140	4.255	2156.65
419	Zion Williamson	2.994	4.401	859.13
167	Anthony Davis	2.983	4.623	1880.81
217	Julius Randle	2.964	4.367	2956.55
16	Derrick White	2.963	4.737	2078.65

	player_name	ridge_rapm	lasso_rapm	minutes
306	Kawhi Leonard	2.927	4.256	1817.22
272	Coby White	2.919	3.688	1795.01
406	Darius Garland	2.896	4.857	2209.82
339	Isaiah Joe	2.800	3.808	1553.19
417	Myles Turner	2.733	2.418	1676.32
243	Jrue Holiday	2.713	2.450	2097.00
260	Michael Porter Jr.	2.661	1.865	1686.43
235	Franz Wagner	2.561	2.162	2293.49

	player_name	lasso_rapm	ridge_rapm	minutes
156	Joel Embiid	9.392	4.992	2249.47
258	Nikola Jokic	8.170	4.310	2255.54
99	Pascal Siakam	6.224	3.738	2497.48
318	Kevin Love	5.715	3.405	1131.82
48	Trae Young	5.193	4.028	2706.40
406	Darius Garland	4.857	2.896	2209.82
16	Derrick White	4.737	2.963	2078.65
167	Anthony Davis	4.623	2.983	1880.81
2	Draymond Green	4.547	3.294	2019.37
419	Zion Williamson	4.401	2.994	859.13
217	Julius Randle	4.367	2.964	2956.55
306	Kawhi Leonard	4.256	2.927	1817.22
242	Brook Lopez	4.255	3.140	2156.65
109	Steven Adams	3.932	2.273	1038.14
89	Harrison Barnes	3.907	2.478	2502.39
339	Isaiah Joe	3.808	2.800	1553.19
272	Coby White	3.688	2.919	1795.01
22	Chris Paul	3.406	2.444	1730.94
218	Jalen Brunson	3.389	3.359	2406.14
26	Cameron Johnson	3.201	2.504	1073.00

	player_name	ridge_rapm	lasso_rapm	ridge_minus_lasso	minutes
142	Devin Vassell	-3.638	-5.744	2.106	1116.12
262	Christian Braun	-2.707	-4.675	1.968	1177.65
215	Anthony Gill	1.681	0.000	1.681	614.33
462	Bogdan Bogdanovic	1.660	0.000	1.660	1516.08
183	Alperen Sengun	1.744	0.125	1.619	1946.20
29	Jock Landale	1.601	0.000	1.601	921.28
429	Alec Burks	1.600	0.004	1.596	1174.32
257	Aaron Gordon	2.208	0.694	1.514	1966.73
103	Scottie Barnes	1.667	0.175	1.492	2553.72
154	James Harden	1.473	0.000	1.473	1974.74

	player_name	ridge_rapm	lasso_rapm	ridge_minus_lasso	minutes
156	Joel Embiid	4.992	9.392	-4.400	2249.47
258	Nikola Jokic	4.310	8.170	-3.860	2255.54

	player_name	ridge_rapm	lasso_rapm	ridge_minus_lasso	minutes
99	Pascal Siakam	3.738	6.224	-2.486	2497.48
318	Kevin Love	3.405	5.715	-2.309	1131.82
320	Robin Lopez	-2.037	-0.000	-2.037	275.70
406	Darius Garland	2.896	4.857	-1.961	2209.82
28	Torrey Craig	-1.901	-0.000	-1.901	1898.98
210	Deni Avdija	-1.957	-0.079	-1.879	1989.40
67	Simone Fontecchio	-1.851	-0.000	-1.851	745.42
223	Obi Toppin	-1.813	-0.000	-1.813	888.69

The broad pattern is that lasso is much sparser than ridge and pushes a smaller set of players to larger absolute values. The top of the lasso list overlaps substantially with the top of the ridge list, but the lasso coefficients are more extreme for the biggest stars and exactly zero for many players that ridge still treats as meaningfully positive or negative.

Interpretation

Looking at the biggest disagreements and then checking how those players looked in the following season, a few patterns stand out.

- On the lasso-greater-than-ridge side, Nikola Jokic and Joel Embiid both clearly validate the idea that true top-end stars can deserve very large coefficients. Jokic won the 2023-24 MVP, and Embiid remained an elite offensive player when healthy. Pascal Siakam also had a strong 2023-24 season and helped Indiana reach the conference finals. Those cases make the lasso's aggressive treatment of a few stars look reasonable.
- On the ridge-greater-than-lasso side, Scottie Barnes, Alperen Sengun, and James Harden all had stronger 2023-24 seasons than a zero-or-near-zero lasso coefficient would suggest. Barnes became an All-Star, Sengun was a Most Improved finalist, and Harden was still a high-level creator for the Clippers. Those cases make ridge look better at preserving strong but less singular impact players.
- Kevin Love is a useful counterexample on the lasso side. Lasso rates him much more highly than ridge, but his 2023-24 role was much smaller, which makes that lasso estimate look too aggressive.

Overall, I would give ridge the edge in the disagreement cases because it seems less likely to overreact to one-season lineup structure and less likely to erase good players who are clearly helpful but not absolute focal points. Lasso does a better job of separating out the most obvious stars, but ridge looks more realistic across the broader set of disagreements.

5.2 Part (b): Interval Caveat

The subtle flaw is that the interval calculations were treated as if the players and model were chosen in advance, but they were not. In practice, I first used the same data to identify the top players and to choose the tuning level, and only after that did I compute intervals. That means the reported intervals ignore selection uncertainty and the winner's-curse effect that comes from focusing on the most extreme estimated coefficients.

Interpretation

The intervals are still useful as rough summaries of uncertainty, but they are too optimistic if interpreted as fully valid post-selection confidence or credible intervals for the "top 20" players. A stricter analysis would account for the fact that the ranking itself and the amount of regularization were learned from the same sample.

6 Conclusion

Overall, the weighted ridge model still looks like the most defensible summary here because it handles lineup collinearity and also gives more influence to longer, more informative stints. The main lesson from the full set of plots and tables is that minutes and uncertainty matter a lot when interpreting any RAPM ranking.